Bird Species Identification from an Image

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UE24CS352A Machine Learning • Project Write-up

1. Problem Statement

Bird species identification is a fine-grained image classification problem: different species share the same basic body parts and differ only in small details such as colour patterns, bill shape and wing shape, while images of a single species vary strongly with pose, lighting, background and occlusion. Our goal is a system in which a user uploads a photograph of a bird and receives the predicted species, a confidence score and the five most likely species. Formally, this is a supervised, single-label, multiclass classification task over 200 species. We build on a Stanford CS 229 project report with the same title (Bhandari, Joshi and Patki), which classified species from human-annotated attributes. Because that approach cannot accept a raw photograph, we learn features directly from the image instead.

2. Dataset

We use the Caltech-UCSD Birds-200-2011 (CUB-200-2011) dataset (Wah et al., 2011): 11,788 images of 200 bird species, collected from Flickr and filtered by Mechanical Turk workers. Our exploratory analysis found 41 to 60 images per species (29 to 30 per species in the official training split), image sides between 120 and 500 pixels, 8 grayscale images and no unreadable files. The classes are therefore nearly balanced. The official training split (5,994 images) was divided into 5,094 training and 900 validation images; the official test split (5,794 images) was kept apart and used only for the final evaluation. All images are resized to 224 x 224 pixels, the input size of the pretrained network.

Split	Images	Purpose
Training	5,094	Fitting the model
Validation	900	Monitoring during training
Test	5,794	Final evaluation only

Table 1. Data split used in this project.

3. Approach

Base paper. The report represents each image by 312 binary attributes (for example, bill shape or back colour) annotated by people, and compares seven scikit-learn classifiers on a random 70/30 split, with PCA and feature selection on the best three. Its best test accuracy is 53.65% (logistic regression with PCA and feature selection); Naive Bayes, KNN, decision trees and random forests reach 19.22%, 31.18%, 24.35% and 33.58% respectively. Decision trees fit the training data almost perfectly (99.83%) but generalise poorly.

Our model. We use transfer learning. A MobileNetV2 network pretrained on ImageNet serves as the feature extractor, with its backbone frozen during initial training. On top of it we add a Global Average Pooling layer, a Dropout layer and a Dense output layer with 200 units and Softmax activation. The model is trained with the Adam optimizer and the sparse categorical cross-entropy loss, which suits integer class labels. Reusing features learned on a large dataset is appropriate here because the training set has only about 25 images per species, too few to train a deep network from scratch.

4. Implementation

The model was implemented in Python with Keras. The trained model is saved as *bird_species_model.keras* and the 200 class names as *bird_species_classes.json*. A Streamlit application loads both files locally: the user uploads an image, which is resized to 224 x 224 and passed to the model, and the application displays the predicted species, its confidence and the top-five predictions. In two application tests the model predicted American Goldfinch with 77.93% confidence and Blue Jay with 95.75% confidence; these are confidences for individual predictions and not a measure of overall accuracy.

5. Results

The model was evaluated once on the 5,794 held-out test images. Top-1 test accuracy is **56.82%** with a test loss of **1.6459**. Table 2 places this next to the base paper's best result. The 3.17 percentage-point difference should be read as indicative only: the paper classifies from annotated attributes on a random 70/30 split, whereas we classify raw images on the official split, so the two protocols are not directly comparable. Top-5 accuracy, macro precision, recall and F1-score, and the most confused species pairs were not measured in this implementation.

Approach	Input	Test accuracy
Base paper: logistic regression + PCA + feature selection	312 attributes	53.65%
This work: MobileNetV2 transfer learning	Raw image	56.82%

Table 2. Test accuracy of the base paper and of this work (different protocols).

6. Conclusion

We built an end-to-end bird species identifier: MobileNetV2 transfer learning on CUB-200-2011 reaches 56.82% top-1 accuracy on 5,794 held-out images across 200 species, and a Streamlit application returns the species, confidence and top-five predictions for an uploaded photograph. Unlike the base paper, the system needs no manually supplied attributes, which is the direction that paper proposed as future work. Limitations are that only top-1 accuracy and loss have been measured, only one model was trained, the comparison with the paper is not like-for-like, and the classifier can predict only the 200 species it was trained on. Future work includes measuring top-5 accuracy, precision, recall, F1-score and a confusion analysis, comparing further backbones and fine-tuning strategies, adding visual explanations such as Grad-CAM, and building a mobile application.

References

[1] A. Bhandari, A. Joshi and R. Patki, *Bird Species Identification from an Image*, CS 229 final project report, Stanford University.

[2] C. Wah, S. Branson, P. Welinder, P. Perona and S. Belongie, *The Caltech-UCSD Birds-200-2011 Dataset*, Technical Report CNS-TR-2011-001, California Institute of Technology, 2011.